



Laxmi Charitable Trust's
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Department of Information Technology (Bsc.IT Semester III)
Python Programming Practical

Practical – 6

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Practical No. 6 - Write Algorithm, Code and Output

a. Write a program to compute number of characters and words in a string.

Code:

Write a program to compute number of Characters and

Words in a string.

'''Use split() method for word count. It divides the string into words using whitespace as the delimiter and len() counts the number of elements in the resulting list. len(s) counts the total number of characters in the string (including letters, spaces and punctuation).'''

s="Second year IT students"

word_count=len(s.split())

```
char_count=len(s)
```

```
#printing result
```

```
print("The number of the words in a string are  
:" +str(word_count))
```

```
print("The number of characters a string are :",char_count)
```

```
print("Himanshu S037")
```

Output:

```
The number of the words in a string are :4  
The number of characters a string are : 23  
Himanshu S037
```

b. Create a file geometry.py to calculate base areas for shapes square and circle. In another file, write a function pointyShapeVolume(x, y, squareBase) that calculates the volume of a square pyramid if squareBase is True and of a right circular cone if squareBase is False. x is the length of an edge on a square if squareBase is True and the radius of a circle when squareBase is False. y is the height of the object. First use squareBase to distinguish the cases. Use the circleArea and squareArea from the geometry module to calculate the base areas.

Code:

```
import geometry
```

```
def pointyShapeVolume(x, y, squareBase):
```

```
    if squareBase:
```

```
        baseArea = geometry.squareArea(x)
```

```
    else:
```

```
        baseArea = geometry.circleArea(x)
```

```
    volume = (1/3) * baseArea * y
```

```
    return volume
```

```
print("Square Pyramid Volume =", pointyShapeVolume(4, 6, True))
```

```
print("Cone Volume =", pointyShapeVolume(3, 5, False))
print("Himanshu S037")
```

```
import math
```

```
def squareArea(x):
    return x**2
```

```
def circleArea(r):
    return math.pi*r**2
```

Output:

```
= RESTART: C:/Users/mvlui/AppData/Local/Programs/Python/Python314/Himanshu/main.py
Square Pyramid Volume = 32.0
Cone Volume = 47.12388980384689
Himanshu S037
```

```
= RESTART: C:/Users/mvlui/AppData/Local/Programs/Python/Python314/Himanshu/geometry.py
```

Q. 2. How can you insert a substring into a string?

Q. 3. List various functions of string with examples.